

Group 'A' (Multiple Choice Questions)

[6 × 1=6]

1. Choose the best alternative and write it in the answer sheet:

a. Which group of fundamental units represents the derived unit of energy joule (J)

- i. kgms^{-2} ii. kgms^{-1} iii. $\text{kgm}^2\text{s}^{-2}$ iv. kgms^{-3}

Answer: iii. kgms^{-2}

b. Which of the following group belongs to Kingdom Plantae?

- i. Fungi ii. Protista iii. Algae iv. Monera

Answer: iii. Algae

c. Which stage of the honey bee is shown in the given figure?

- i. Egg ii. Larva iii. Pupa iv. Adult

Answer: iii. Pupa



d. What is the weight of a body on the surface of the Moon if its weight is 150 N on the Earth's surface?

- i. 20 N ii. 25 N iii. 30 N iv. 35 N

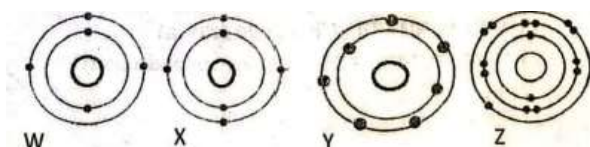
Answer: ii. 25 N

e. In which of the following cases is the total number of electrons 10?

- i. O^- ii. Mg^+ iii. Na^+ iv. N^{--}

Answer: iii. Na^+

f. Which of the following non-metals is more reactive?



- i. W ii. X iii. Y iv. Z

Answer: iii. Y

Group 'B' (Very Short Answer Questions)

[6 × 1=6]

2. Answer the following questions:

a. Derive the fundamental units involved in the unit of power.

$$\text{Answer: Power (P)} = \frac{\text{Work (W)}}{\text{Time taken (t)}} = \frac{\text{J}}{\text{s}}$$

$$\text{Fundamental units in Power (P)} = \frac{\text{kg m} \times \text{m}}{\text{s}^2 \times \text{s}} \quad [\because \text{J} = \text{Nm}]$$

$$\text{or, P} = \frac{\text{kgm}^2}{\text{s}^3} \quad \therefore \text{Pa} = \text{kg m}^2/\text{s}^3$$

b. Although fern is a non-flowering plant, it is classified along with flowering plants in the division Tracheophyta. Give a reason.

Answer: Ferns are classified in Tracheophyta because, like flowering plants, they possess a well-developed vascular system (xylema and phloem) for conducting water and minerals.

c. Write any one characteristic of a queen bee.

Answer: Queen bee has small rounded head and short proboscis covered with hairs.

d. Write any one difference between mass and weight.

Mass		Weight	
i.	The total matter contained in a body is called mass.	i.	The measurement of force of attraction with which planet or satellite attracts anybody towards its center is called weight of body

e. What do you mean by "the acceleration due to gravity of an object is 9.8 m/s²?"

Answer: It means that the velocity of a freely falling object increases at a rate of 9.8 m/s every second.

f. Write the Modern Periodic Law.

Answer: "The physical and chemical properties of elements are the periodic functions of their atomic numbers".

Group 'C' (Short Answer Questions)

[9 × 2 = 18]

3. Test the validity of the $s = ut + \frac{1}{2}at^2$ equation using unit analysis.

Solution : Given, $s = ut + \frac{1}{2}at^2$

Taking unit on both sides.

$$\text{or, } m = \frac{m}{s} \times s + \frac{m^2}{s^4} \times s \quad \text{or, } m = m^2 + \frac{m^2}{s^3}$$

or, Unit of LHS $\neq$ Unit of RHS

Hence, given equation is heterogeneous and invalid.

4. Write any two characteristics of Angiosperms.

Answer: i. Their seeds are enclosed within fruits.

ii. They have a highly developed vascular system.

5. Why are the given organisms (Pigeon and Bat) placed in different classes?

Explain on the basis of reproduction.

Answer: Although they both fly, they belong to different classes based on reproduction: the Pigeon (Aves) is oviparous (lays eggs), while the Bat (Mammalia) is viviparous (gives birth to live young) and has mammary glands to feed milk.



6. Which insect would you suggest poor villagers cultivate to improve their economic condition? Give two reasons.

Answer: I would suggest cultivate Honey bees due to following reasons:

i. Bee-keeping provides income through the sale of honey and wax.

ii. Bees act as pollinators, which significantly increases the yield of agricultural crops and flowers.

7. Write one function of worker bees during 7–11 days and 18–20 days.

Answer: **7–11 days:** They produce royal jelly and feed it to the larvae.

18–20 days: They guard the entrance of the hive.

8. Hari and Ramesh sell apples from Mustang in Butwal. Hari used a beam balance while Ramesh used a spring balance. Who would make more profit? Why?

Answer: Hari would make more profit. A beam balance measures mass, which remains constant regardless of altitude. A spring balance measures weight, which is lower in Mustang (higher altitude) due to less acceleration due to gravity. Ramesh would buy a larger quantity of apples for the same "weight" in Mustang, but sell fewer by "weight" in Butwal where acceleration due to gravity is more, leading to a loss in quantity.

9. The mass and radius of Jupiter are 1.9×10^{27} kg and 71×10^6 m respectively, find the acceleration due to gravity of it. What will be the weight of an object having mass 80 kg on that planet?

Solution: Given, Mass of Jupiter (M) = 1.9×10^{27} kg
 Radius of Jupiter (R) = 71×10^6 m
 Acceleration due to gravity (g) = ?, Weight of an object = ?

Using formula, $g = \frac{GM}{R^2} = \frac{6.67 \times 10^{-11} \times 1.9 \times 10^{27}}{(71 \times 10^6)^2}$ or, $g = 25.14 \text{ m/s}^2$

Again, $W = mg$ or, $W = 80 \times 25.14 = 2011.2 \text{ N}$ $\therefore$ Weight of an object = 2011.2 N

10. Prove that if the Earth attracts two bodies at the same distance from its center with the same force, their masses are equal.

Answer: Let masses of two bodies are m_1 and m_2 respectively and they are kept at same distance d from the center of the earth.

Gravitational force between earth and first body $F = \frac{GMm_1}{d^2}$ (i)

Gravitational force between earth and first body $F = \frac{GMm_2}{d^2}$ (ii)

From (i) and (ii) we get,

$\frac{GMm_1}{d^2} = \frac{GMm_2}{d^2}$ (ii)

$m_1 = m_2$ Therefore, masses are equal.



11. Write any two differences between s-block and p-block elements.

s-block element		p-block element	
i.	The elements with the last electron entering in the s-subshell are called s-block elements.	i.	The elements with the last electron entering in the p-subshell are called p-block elements.
ii.	s-block consists only of metals.	ii.	p-block contains metals, non-metals, and noble gases.

Group 'D' (Long Answer Questions)

[5×4 = 20]

12. Find independent and dependent variables in following events.

i. More growth of plant in Sunlight:

Answer : Independent: Sunlight and Dependent: Plant growth.

ii. Measuring the stretch of a catapult and the distance covered by the stone thrown by it.

Answer : Independent: Stretch of catapult and Dependent: Distance covered by the stone.



13. Write two similarities and differences between gymnosperms and angiosperms.

Answer: Similarities between gymnosperms and angiosperms.

Gymnosperms		Angiosperms	
i.	They are flowering plants.	i.	They are flowering plants.
ii.	They have have well-developed vascular tissues (xylem and phloem).	ii.	They have have well-developed vascular tissues (xylem and phloem).

Differences between gymnosperms and angiosperms.

Gymnosperms	Angiosperms
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i.	They have naked seeds.	i.	They have seeds enclosed in fruits.
ii.	They have no ovary in flower and produce cones.	ii.	They ovary in flower and produce flowers.

14. Answer the following questions on the basis of given figure.

i. Label A, B, C, and D:

Answer : A: Egg, B: Larva, C: Pupa, D: Adult.

ii. What is the process called when stage B sheds its old skin?

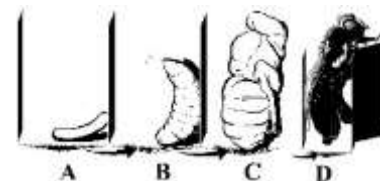
Answer : The process when B sheds its old skin is called Moulting.

iii. Under what condition does B develop into a worker bee?

It must be a fertilized egg fed with royal jelly for only the first 3 days.

iv. How long does B take to form its next stage?

The larval stage (B) takes about 5 to 6 days to become a pupa (C).



15. a. If a body is dropped from the same height once at the equator and then at the polar region, in which place will it fall faster? Explain with reason.

Radius of the earth is less in polar region than in equatorial region. The value of acceleration due to gravity is inversely proportional to the square of the radius of the earth. Hence, the value of acceleration due to gravity is more in the polar region than the equatorial region. The body will fall faster in polar region.

b. Write any two differences between acceleration due to gravity and universal gravitational constant.

Acceleration due to gravity (g)		Universal Gravitational constant (G)	
i.	The acceleration produced in a freely falling body due to gravity is called acceleration due to gravity.	i.	The force of attraction between two bodies of unit masses placed at unit distance apart from their centers is called universal gravitational constant.
ii.	The value of g changes from place to place.	ii.	The value of G remains constant everywhere.

16. A part of periodic table is shown below in which some hypothetical symbols are used to represent real elements. Study the table and answer the following questions.

Period	IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA or 0
n=2	C	B						Ne
n=3							A	D

i. Write the electronic configuration of element 'A' on the basis of sub-shell.

A : $1s^2, 2s^2 2p^6, 3s^2 3p^5$

ii. Element 'C' is more reactive than 'B', why?

The element C (Li) has one valence electrons and the element B (Be) has two valence electrons. It is more easily to lose one electron than two electrons. Hence, the element C (Li) is more reactive between B and C.

iii. Atomic size of 'A' is greater than element 'D', why ?

As the number of proton is more with element 'D', the nuclear attraction between electron and nucleus is also more with element 'D' than element 'A'. So, the atomic size of 'D' shrinks and makes the size smaller than element 'A'.

Best of Luck